In summary, four parameters, a_i, α_i, d_i, and θ_i, are associated with each link of a manipulator. If a sign convention for each of these parameters has been established, then these parameters constitute a sufficient set to completely determine the kinematic configuration of each link of a robot arm. Note that these four parameters come in pairs: the link parameters (a_i, α_i) which determine the structure of the link and the joint parameters (d_i, θ_i) which determine the relative position of neighboring links.

2.2.10 The Denavit-Hartenberg Representation

To describe the translational and rotational relationships between adjacent links, Denavit and Hartenberg [1955] proposed a matrix method of systematically establishing a coordinate system (body-attached frame) to each link of an articulated chain. The Denavit-Hartenberg (D-H) representation results in a 4×4 homogeneous transformation matrix representing each link's coordinate system at the joint with respect to the previous link's coordinate system. Thus, through sequential transformations, the end-effector expressed in the "hand coordinates" can be transformed and expressed in the "base coordinates" which make up the inertial frame of this dynamic system.

An orthonormal cartesian coordinate system $(\mathbf{x}_i, \mathbf{y}_i, \mathbf{z}_i)$† can be established for each link at its joint axis, where $i = 1, 2, \ldots, n$ (n = number of degrees of freedom) plus the base coordinate frame. Since a rotary joint has only 1 degree of freedom, each $(\mathbf{x}_i, \mathbf{y}_i, \mathbf{z}_i)$ coordinate frame of a robot arm corresponds to joint $i + 1$ and is fixed in link i. When the joint actuator activates joint i, link i will move with respect to link $i - 1$. Since the ith coordinate system is fixed in link i, it moves together with the link i. Thus, the nth coordinate frame moves with the hand (link n). The base coordinates are defined as the 0th coordinate frame $(\mathbf{x}_0, \mathbf{y}_0, \mathbf{z}_0)$ which is also the inertial coordinate frame of the robot arm. Thus, for a six-axis PUMA-like robot arm, we have seven coordinate frames, namely, $(\mathbf{x}_0, \mathbf{y}_0, \mathbf{z}_0), (\mathbf{x}_1, \mathbf{y}_1, \mathbf{z}_1), \ldots, (\mathbf{x}_6, \mathbf{y}_6, \mathbf{z}_6)$.

Every coordinate frame is determined and established on the basis of three rules:

1. The $\mathbf{z}_{i-1}$ axis lies along the axis of motion of the ith joint.
2. The $\mathbf{x}_i$ axis is normal to the $\mathbf{z}_{i-1}$ axis, and pointing away from it.
3. The $\mathbf{y}_i$ axis completes the right-handed coordinate system as required.

By these rules, one is free to choose the location of coordinate frame 0 anywhere in the supporting base, as long as the $\mathbf{z}_0$ axis lies along the axis of motion of the first joint. The last coordinate frame (nth frame) can be placed anywhere in the hand, as long as the $\mathbf{x}_n$ axis is normal to the $\mathbf{z}_{n-1}$ axis.

The D-H representation of a rigid link depends on four geometric parameters associated with each link. These four parameters completely describe any revolute

† $(\mathbf{x}_i, \mathbf{y}_i, \mathbf{z}_i)$ actually represent the unit vectors along the principal axes of the coordinate frame i, respectively, but are used here to denote the coordinate frame i.

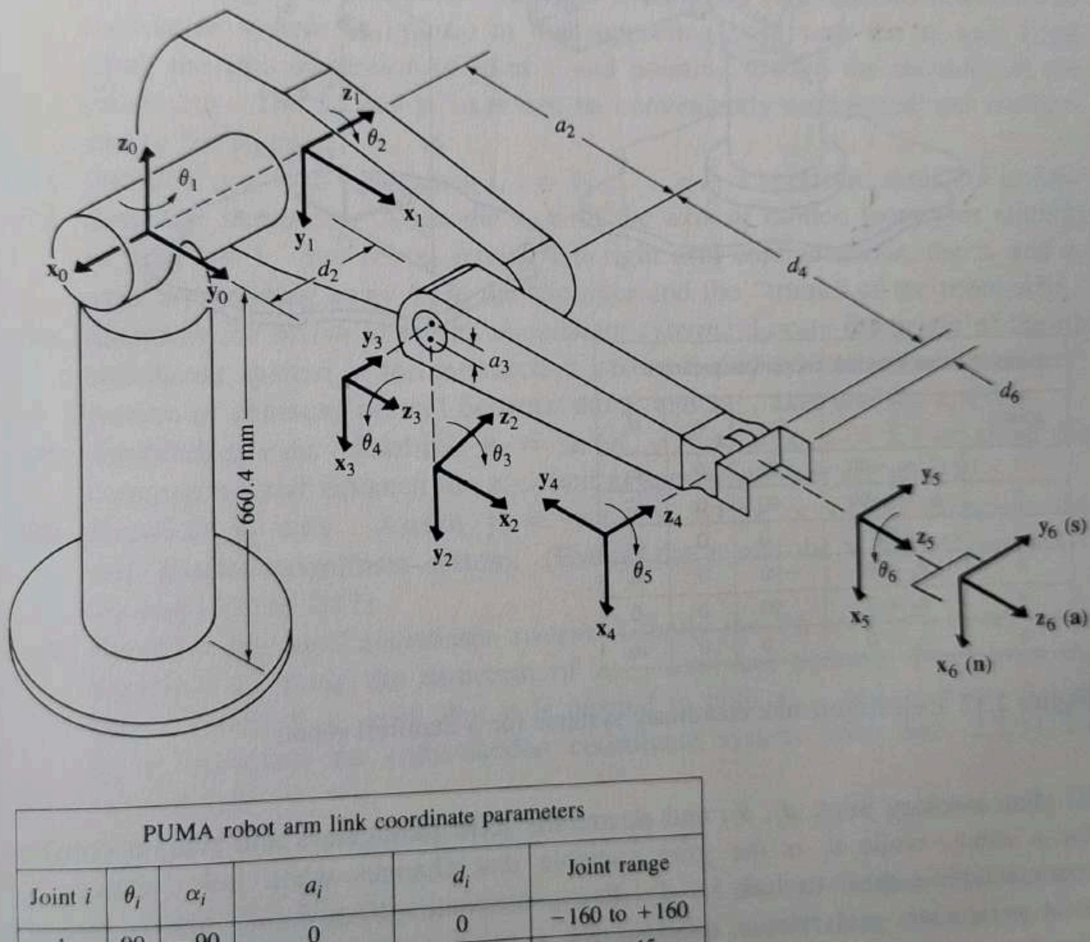
or prismatic joint. Referring to Fig. 2.10, these four parameters are defined as follows:

θ_i is the joint angle from the $\mathbf{x}_{i-1}$ axis to the $\mathbf{x}_i$ axis about the $\mathbf{z}_{i-1}$ axis (using the right-hand rule).

d_i is the distance from the origin of the $(i-1)$th coordinate frame to the intersection of the $\mathbf{z}_{i-1}$ axis with the $\mathbf{x}_i$ axis along the $\mathbf{z}_{i-1}$ axis.

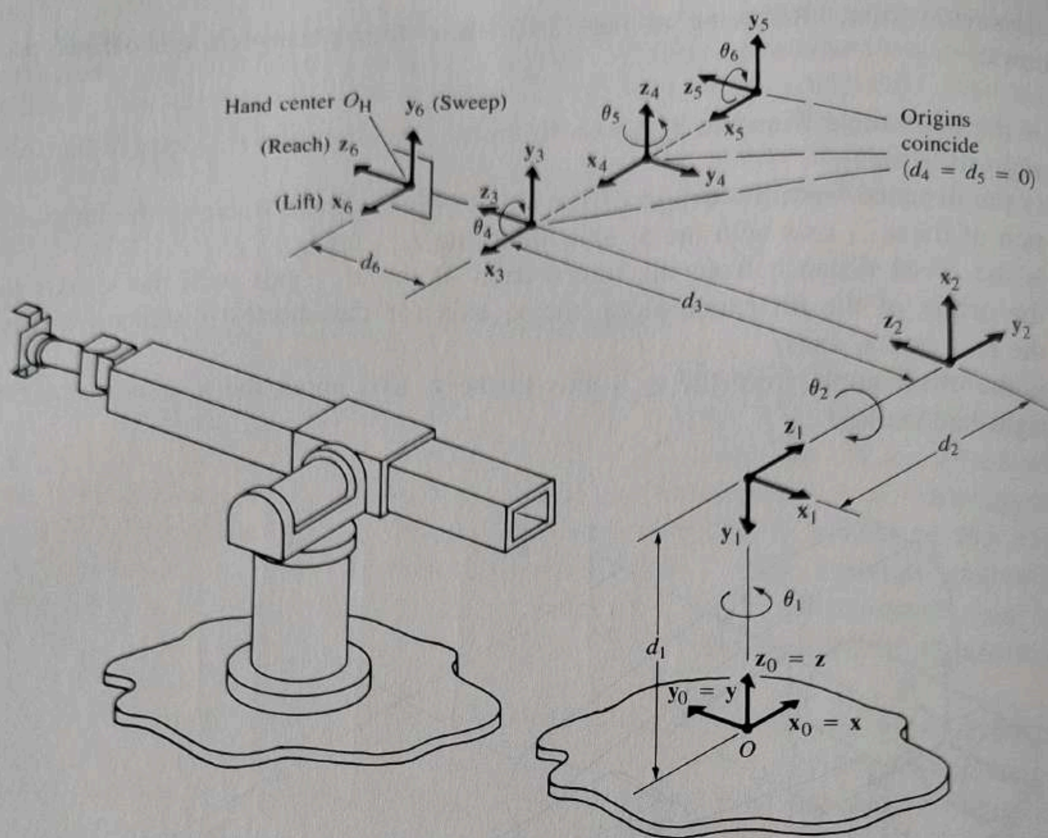
a_i is the offset distance from the intersection of the $\mathbf{z}_{i-1}$ axis with the $\mathbf{x}_i$ axis to the origin of the ith frame along the $\mathbf{x}_i$ axis (or the shortest distance between the $\mathbf{z}_{i-1}$ and $\mathbf{z}_i$ axes).

α_i is the offset angle from the $\mathbf{z}_{i-1}$ axis to the $\mathbf{z}_i$ axis about the $\mathbf{x}_i$ axis (using the right-hand rule).



PUMA robot arm link coordinate parameters					
Joint i	θ_i	α_i	a_i	d_i	Joint range
1	90	−90	0	0	−160 to +160
2	0	0	431.8 mm	149.09 mm	−225 to 45
3	90	90	−20.32 mm	0	−45 to 225
4	0	−90	0	433.07 mm	−110 to 170
5	0	90	0	0	−100 to 100
6	0	0	0	56.25 mm	−266 to 266

Figure 2.11 Establishing link coordinate systems for a PUMA robot.



Stanford robot arm link coordinate parameters				
Joint i	θ_i	α_i	a_i	d_i
1	$\theta_1 = -90$	-90	0	d_1
2	$\theta_2 = -90$	90	0	d_2
3	-90	0	0	d_3
4	$\theta_4 = 0$	-90	0	0
5	$\theta_5 = 0$	90	0	0
6	$\theta_6 = 0$	0	0	d_6

Figure 2.12 Establishing link coordinate systems for a Stanford robot.

For a rotary joint, d_i, a_i, and α_i are the joint parameters and remain constant for a robot, while θ_i is the joint variable that changes when link i moves (or rotates) with respect to link $i - 1$. For a prismatic joint, θ_i, a_i, and α_i are the joint parameters and remain constant for a robot, while d_i is the joint variable. For the remainder of this book, *joint variable* refers to θ_i (or d_i), that is, the varying quantity, and *joint parameters* refer to the remaining three geometric constant values (d_i, a_i, α_i) for a rotary joint, or (θ_i, a_i, α_i) for a prismatic joint.

With the above three basic rules for establishing an orthonormal coordinate system for each link and the geometric interpretation of the joint and link parameters, a procedure for establishing *consistent* orthonormal coordinate systems for a robot is outlined in Algorithm 2.1. Examples of applying this algorithm to a six-axis PUMA-like robot arm and a Stanford arm are given in Figs. 2.11 and 2.12, respectively.

Algorithm 2.1: Link Coordinate System Assignment. Given an n degree of freedom robot arm, this algorithm assigns an orthonormal coordinate system to each link of the robot arm according to arm configurations similar to those of human arm geometry. The labeling of the coordinate systems begins from the supporting base to the end-effector of the robot arm. The relations between adjacent links can be represented by a 4×4 homogeneous transformation matrix. The significance of this assignment is that it will aid the development of a consistent procedure for deriving the joint solution as discussed in the later sections. (Note that the assignment of coordinate systems is not unique.)

D1. *Establish the base coordinate system.* Establish a right-handed orthonormal coordinate system $(\mathbf{x}_0, \mathbf{y}_0, \mathbf{z}_0)$ at the supporting base with the $\mathbf{z}_0$ axis lying along the axis of motion of joint 1 and pointing toward the shoulder of the robot arm. The $\mathbf{x}_0$ and $\mathbf{y}_0$ axes can be conveniently established and are normal to the $\mathbf{z}_0$ axis.

D2. *Initialize and loop.* For each i, $i = 1, \ldots, n-1$, perform steps D3 to D6.

D3. *Establish joint axis.* Align the $\mathbf{z}_i$ with the axis of motion (rotary or sliding) of joint $i + 1$. For robots having left-right arm configurations, the $\mathbf{z}_1$ and $\mathbf{z}_2$ axes are pointing away from the shoulder and the "trunk" of the robot arm.

D4. *Establish the origin of the* ith *coordinate system.* Locate the origin of the ith coordinate system at the intersection of the $\mathbf{z}_i$ and $\mathbf{z}_{i-1}$ axes or at the intersection of common normal between the $\mathbf{z}_i$ and $\mathbf{z}_{i-1}$ axes and the $\mathbf{z}_i$ axis.

D5. *Establish* $\mathbf{x}_i$ *axis.* Establish $\mathbf{x}_i = \pm (\mathbf{z}_{i-1} \times \mathbf{z}_i)/\|\mathbf{z}_{i-1} \times \mathbf{z}_i\|$ or along the common normal between the $\mathbf{z}_{i-1}$ and $\mathbf{z}_i$ axes when they are parallel.

D6. *Establish* $\mathbf{y}_i$ *axis.* Assign $\mathbf{y}_i = +(\mathbf{z}_i \times \mathbf{x}_i)/\|\mathbf{z}_i \times \mathbf{x}_i\|$ to complete the right-handed coordinate system. (Extend the $\mathbf{z}_i$ and the $\mathbf{x}_i$ axes if necessary for steps D9 to D12).

D7. *Establish the hand coordinate system.* Usually the nth joint is a rotary joint. Establish $\mathbf{z}_n$ along the direction of $\mathbf{z}_{n-1}$ axis and pointing away from the robot. Establish $\mathbf{x}_n$ such that it is normal to both $\mathbf{z}_{n-1}$ and $\mathbf{z}_n$ axes. Assign $\mathbf{y}_n$ to complete the right-handed coordinate system. (See Sec. 2.2.11 for more detail.)

D8. *Find joint and link parameters.* For each i, $i = 1, \ldots, n$, perform steps D9 to D12.

D9. *Find* d_i. d_i is the distance from the origin of the $(i-1)$th coordinate system to the intersection of the $\mathbf{z}_{i-1}$ axis and the $\mathbf{x}_i$ axis along the $\mathbf{z}_{i-1}$ axis. It is the joint variable if joint i is prismatic.

D10. *Find* a_i. a_i is the distance from the intersection of the $\mathbf{z}_{i-1}$ axis and the $\mathbf{x}_i$ axis to the origin of the ith coordinate system along the $\mathbf{x}_i$ axis.

D11. *Find* θ_i. θ_i is the angle of rotation from the $\mathbf{x}_{i-1}$ axis to the $\mathbf{x}_i$ axis about the $\mathbf{z}_{i-1}$ axis. It is the joint variable if joint i is rotary.

D12. *Find* α_i. α_i is the angle of rotation from the $\mathbf{z}_{i-1}$ axis to the $\mathbf{z}_i$ axis about the $\mathbf{x}_i$ axis.

Once the D-H coordinate system has been established for each link, a homogeneous transformation matrix can easily be developed relating the ith coordinate frame to the $(i-1)$th coordinate frame. Looking at Fig. 2.10, it is obvious that a point $\mathbf{r}_i$ expressed in the ith coordinate system may be expressed in the $(i-1)$th coordinate system as $\mathbf{r}_{i-1}$ by performing the following successive transformations:

1. Rotate about the $\mathbf{z}_{i-1}$ axis an angle of θ_i to align the $\mathbf{x}_{i-1}$ axis with the $\mathbf{x}_i$ axis ($\mathbf{x}_{i-1}$ axis is parallel to $\mathbf{x}_i$ and pointing in the same direction).
2. Translate along the $\mathbf{z}_{i-1}$ axis a distance of d_i to bring the $\mathbf{x}_{i-1}$ and $\mathbf{x}_i$ axes into coincidence.
3. Translate along the $\mathbf{x}_i$ axis a distance of a_i to bring the two origins as well as the $\mathbf{x}$ axis into coincidence.
4. Rotate about the $\mathbf{x}_i$ axis an angle of α_i to bring the two coordinate systems into coincidence.

Each of these four operations can be expressed by a basic homogeneous rotation-translation matrix and the product of these four basic homogeneous transformation matrices yields a composite homogeneous transformation matrix, ${}^{i-1}\mathbf{A}_i$, known as the D-H transformation matrix for adjacent coordinate frames, i and $i-1$. Thus,

$$
\begin{aligned}
{}^{i-1}\mathbf{A}_i &= \mathbf{T}_{z,d}\,\mathbf{T}_{z,\theta}\,\mathbf{T}_{x,a}\,\mathbf{T}_{x,\alpha} \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & 0 \\ \sin\theta_i & \cos\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha_i & -\sin\alpha_i & 0 \\ 0 & \sin\alpha_i & \cos\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} \cos\theta_i & -\cos\alpha_i\,\sin\theta_i & \sin\alpha_i\,\sin\theta_i & a_i\,\cos\theta_i \\ \sin\theta_i & \cos\alpha_i\,\cos\theta_i & -\sin\alpha_i\,\cos\theta_i & a_i\,\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
\end{aligned}
\tag{2.2-29}
$$

Using Eq. (2.2-27), the inverse of this transformation can be found to be

$$
[{}^{i-1}\mathbf{A}_i]^{-1} = {}^{i}\mathbf{A}_{i-1} = \begin{bmatrix} \cos\theta_i & \sin\theta_i & 0 & -a_i \\ -\cos\alpha_i\sin\theta_i & \cos\alpha_i\,\cos\theta_i & \sin\alpha_i & -d_i\,\sin\alpha_i \\ \sin\alpha_i\sin\theta_i & -\sin\alpha_i\cos\theta_i & \cos\alpha_i & -d_i\cos\alpha_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
\tag{2.2-30}
$$

where α_i, a_i, d_i are constants while θ_i is the joint variable for a revolute joint.

For a prismatic joint, the joint variable is d_i, while α_i, a_i, and θ_i are constants. In this case, ${}^{i-1}\mathbf{A}_i$ becomes

$$ {}^{i-1}\mathbf{A}_i = \mathbf{T}_{z,\theta}\,\mathbf{T}_{z,d}\,\mathbf{T}_{x,\alpha} = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i \sin\theta_i & \sin\alpha_i \sin\theta_i & 0 \\ \sin\theta_i & \cos\alpha_i \cos\theta_i & -\sin\alpha_i \cos\theta_i & 0 \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.2-31} $$

and its inverse is

$$ [{}^{i-1}\mathbf{A}_i]^{-1} = {}^{i}\mathbf{A}_{i-1} = \begin{bmatrix} \cos\theta_i & \sin\theta_i & 0 & 0 \\ -\cos\alpha_i \sin\theta_i & \cos\alpha_i \cos\theta_i & \sin\alpha_i & -d_i \sin\alpha_i \\ \cos\alpha_i \sin\theta_i & -\sin\alpha_i \cos\theta_i & \cos\alpha_i & -d_i \cos\alpha_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.2-32} $$

Using the ${}^{i-1}\mathbf{A}_i$ matrix, one can relate a point $\mathbf{p}_i$ at rest in link i, and expressed in homogeneous coordinates with respect to coordinate system i, to the coordinate system $i - 1$ established at link $i - 1$ by

$$ \mathbf{p}_{i-1} = {}^{i-1}\mathbf{A}_i\, \mathbf{p}_i \tag{2.2-33} $$

where $\mathbf{p}_{i-1} = (x_{i-1}, y_{i-1}, z_{i-1}, 1)^T$ and $\mathbf{p}_i = (x_i, y_i, z_i, 1)^T$.

The six ${}^{i-1}\mathbf{A}_i$ transformation matrices for the six-axis PUMA robot arm have been found on the basis of the coordinate systems established in Fig. 2.11. These ${}^{i-1}\mathbf{A}_i$ matrices are listed in Fig. 2.13.

2.2.11 Kinematic Equations for Manipulators

The homogeneous matrix ${}^0\mathbf{T}_i$ which specifies the location of the ith coordinate frame with respect to the base coordinate system is the chain product of successive coordinate transformation matrices of ${}^{i-1}\mathbf{A}_i$, and is expressed as

$$ {}^0\mathbf{T}_i = {}^0\mathbf{A}_1\, {}^1\mathbf{A}_2 \cdots {}^{i-1}\mathbf{A}_i = \prod_{j=1}^{i} {}^{j-1}\mathbf{A}_j \qquad \text{for } i = 1, 2, \ldots, n $$

$$ = \begin{bmatrix} \mathbf{x}_i & \mathbf{y}_i & \mathbf{z}_i & \mathbf{p}_i \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} {}^0\mathbf{R}_i & {}^0\mathbf{p}_i \\ 0 & 1 \end{bmatrix} \tag{2.2-34} $$

where

$[\mathbf{x}_i, \mathbf{y}_i, \mathbf{z}_i]$ = orientation matrix of the ith coordinate system established at link i with respect to the base coordinate system. It is the upper left 3×3 partitioned matrix of ${}^0\mathbf{T}_i$.

$\mathbf{p}_i$ = position vector which points from the origin of the base coordinate system to the origin of the ith coordinate system. It is the right 3×1 partitioned matrix of ${}^0\mathbf{T}_i$.